

HW: 13/07/26

1) write a C++ code which includes a class student. Find out the space occupied by the object by using `sizeof()`.

Algorithm Efficiency:-

- * The systematic study of the fundamental techniques used to design and analyze efficient algorithms. $f(n)$ = efficiency
- * Brassard and Bratley coined the term algorithm which they define as:
"the systematic study of the fundamental

technique used to design and analysis efficient algorithms".

$f(n)$ = Efficiency

① Given an array and integer target value find indices of the two numbers in the array whose sum is equal to given target value.

source code:-

```
#include <iostream>
using namespace std;
int main {
    int n, target;
    cout << "Enter the values";
    cin >> n;
    int a[n];
    for (int i = 0; i < n; i++)
    { cin >> a[i];
    }
    cout << "Enter the target value";
    cin >> target;
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] + a[j] == target) {
                cout << i << " " << j << endl;
                return 0;
            }
        }
    }
}
```

ex:- `int sum (int A[], int n)`

{

`int sum = 0, i // "1"`

`for (i = 0; i < n; i++)`

`sum = sum + A[i]`

`return sum;`

}

1	1	1
1+1+1	++ (n+1)++	2n+2
2	n	2n
1	1	3